



Generative AI as an Interface Between Farmers and AI Algorithms: Opportunities and Challenges

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Abstract | Artificial Intelligence (AI) has transformed industries worldwide, yet its adoption in agriculture remains limited, particularly in low-resource settings. Farmers in developing countries face two chronic challenges: (1) the assumption of reliable Internet access is often an incorrect one to make in low-resource settings in developing countries; and (2) the complexity of AI algorithms often makes it challenging to be used by low-resource farmers, who often lack the required significant technical expertise needed to use these AI algorithms effectively. As a result, many low-resource farmers are excluded from the benefits of AI technologies, thereby exacerbating the global digital divide. This paper explores whether Generative AI, particularly large language models (LLMs) like ChatGPT, Gemini, and LLaMA, can bridge this ever-increasing divide, and make AI tools and algorithms more accessible to farmers in developing countries. LLMs represent a breakthrough in interactivity, allowing users to engage conversationally with AI algorithms. Historically, AI has lacked this interactivity, making it difficult for non-technical users to understand or act on algorithmic outputs. Generative AI models, however, offer a solution by serving as intuitive interfaces that translate complex AI algorithms into actionable insights. These models enable farmers to ask questions and receive answers in natural language, along with enabling them to provide feedback so that the underlying AI algorithms could be adapted to their specific needs. Furthermore, by leveraging non-Internet modalities like voice-based interfaces and SMS, these models can extend AI's reach to remote areas lacking reliable Internet access. This paper takes the position that Generative AI can indeed serve as a bridge between low-resource farmers and heavy-duty AI algorithms. This paper discusses how LLMs bring a unique level of interactivity to AI. We then describe a vision for how these models can enable farmers to access algorithmic agricultural advice in an accessible and user-friendly manner. Subsequently, we outline the challenges that must be overcome to realize this vision, including technical, infrastructural, and ethical considerations. Finally, we conclude with a discussion on the broader implications of using Generative AI to address the needs of low-resource farming communities. By exploring this intersection of Generative AI and agriculture, this paper aims to contribute to the ongoing dialogue on making AI technologies inclusive and impactful for underserved populations, ensuring that the transformative potential of AI is enjoyed equitably by all sections of society.

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1 Introduction

Artificial Intelligence (AI) has become an integral part of our daily lives, transforming the way we work, communicate, and solve problems. Often referred to as the second coming of the industrial revolution, the field of AI has revolutionized a wide variety of domains.^{1,2} To name a few domains, AI has been used successfully for improving patient outcomes in healthcare,^{3,4} optimizing logistics in supply chains,⁵ and personalizing user experiences in entertainment and retail,⁶ etc.

The field of AI has also made significant strides in the field of agriculture. For example, AI-powered analysis of remote sensing data from satellite-based products (such as MODIS⁷ and Landsat⁸) enables accurate forecasts about water stress and crop productivity, potentially helping farmers optimize their cropping resources more efficiently.⁹ Similarly, crowdsourced reports from farmers combined with machine learning algorithms have been used to predict biotic stressors¹⁰ such as locust outbreaks, enabling timely interventions. At the same time, wide and deep neural networks have been used to forecast produce prices,¹¹ which can be used by farmers to strategically decide when and where to sell their harvested crops to maximize returns. However, despite such promising advancements, the real-world deployment of these (and other such) AI algorithms in the agriculture domain with real farmers has been limited, particularly in developing countries around the world. Many of these applications remain confined to research settings, with minimal impact on the everyday lives of farmers. This gap arises primarily because AI in agriculture faces two chronic challenges, leaving countless low-resource farmers around the globe excluded from its potential benefits.

The first challenge would be the assumption of Internet access. Most popular AI tools and services are built on top of the Internet layer, operating under the assumption that users will have access to reliable, high-speed Internet. While this assumption may hold true in many industrialized Western nations, it is far from universal. According to the World Bank, as of 2023, 33% of the world's population had never accessed the Internet.¹² This lack of connectivity disproportionately affects rural communities, including farmers in developing countries, leaving them unable to benefit from AI-driven solutions that require robust Internet access. This has led to AI algorithms exacerbating an already-wide digital divide between citizens of the developed vs developing world.

The second challenge would be the lack of interpretability and actionability owing to the black-box nature of most heavy-duty AI algorithms, which hinders accessibility and usability of these algorithms by farmer populations. Even if AI services could be delivered universally (i.e., if Internet access was not an issue), several associated challenges prevent low-resource farmer communities from fully benefiting from AI. One significant issue is that AI algorithms are often predictive rather than prescriptive. For instance, machine learning models might forecast that certain farms are likely to experience a locust infestation in the coming week.¹³ However, these predictions do not directly translate into actionable interventions for farmers to minimize damage. While there has been some progress in decision-focused learning,¹⁴ these techniques are highly technical and remain far removed from practical, real-world applications.

Furthermore, even when AI tools provide prescriptive solutions, understanding and applying these outputs often requires a level of technical knowledge that is beyond the reach of most low-resource farmers. Many of these individuals (especially those belonging to developing countries) may lack access to formal education, and explaining complex concepts like model coefficients, parameters, or eigenvalues becomes nearly impossible. This disconnect leaves farming communities unable to interpret or trust AI-generated recommendations,¹⁵ further limiting the real-world impact of these technologies in resource-constrained settings.

As a result of these challenges, individuals from low-resource communities are often excluded from the benefits of AI technologies, further widening the existing digital divide. This exclusion is particularly concerning because AI was envisioned as a transformative technology capable of uplifting people and enabling them to lead better lives. For many low-resource populations (including smallholder farmers around the world), however, it is doing the opposite—reinforcing inequalities rather than alleviating them. This stark disparity highlights an urgent need to rethink how AI can be made inclusive and accessible. Ensuring that AI addresses the problems faced by low-resource communities around the world is not just a matter of equity, but a critical issue that must be tackled to fulfil AI's true potential as a tool for global progress.

Can Generative AI Help? Instruction-tuned models, the most prominent and widely

recognized manifestation of Generative AI, represent a significant leap in the field of AI. In a nutshell, while Large Language Models (LLMs) are neural network models which are designed for the sole task of next token prediction (i.e., given a sentence prefix, predict the next word in sequence),¹⁶ instruction-tuned models are fine-tuned versions of LLMs that are trained on conversational datasets, so that they can follow instructions provided by users, enabling them to generate coherent and context-aware responses. By tailoring their outputs to specific prompts, instruction-tuned models allow users to interact naturally (via chat) with AI systems, making them an intuitive tool for a variety of applications.

Over the last couple of years, LLMs have taken the world by storm, showcasing human-like capabilities across a wide variety of contexts.¹⁷ In fact, the most popular LLMs such as ChatGPT, Gemini, LLaMA¹⁸, etc., are effectively instruction-tuned models. However, this remarkable success is not without its caveats. A growing body of research has shed light on the limitations of these models, including their susceptibility to generating biased or discriminatory responses, propagating misinformation, and producing potentially harmful outputs in domains such as medicine.^{19–21} These findings underscore the importance of tempering the excitement surrounding Generative AI with a clear understanding of its limitations.

However, in this paper, we take a different standpoint: we explore whether instruction-tuned models (and Generative AI, more broadly) can be the catalyst needed to revolutionize AI for Agriculture. Could it address the longstanding challenges that have hindered the adoption of AI tools and services in this domain? Specifically, we examine whether instruction-tuned models can serve as an interactive interface between farmers (many of whom lack technical knowledge or access to the Internet) and the sophisticated, computationally intensive AI algorithms that require substantial resources to train and deploy.

The rest of this paper is organized as follows. In Sect. 2, we review prior research related to utilizing AI for agriculture problems. In Sect. 3, we present our case for how the interactivity offered by instruction-tuned models is unique, and it should be leveraged to communicate the processing of heavy-duty AI algorithms to non-tech savvy farmers. Section 4 outlines a simple yet practical vision for how Large Language Models (LLMs) could empower farmers by providing

them with access to algorithmic agricultural advice. In Sect. 5, we identify and discuss several key challenges that must be addressed to turn this vision into reality. Finally, Sect. 6 concludes the paper with a broader discussion on the implications of this work and directions for future research.

2 Related Work

Research in AI for agriculture has gained a lot of attention in the past few years. Prior work can be divided into three directions: (1) machine learning (ML) approaches applied to the agricultural domain; and (2) the recently developed Generative AI applications for global agriculture; (3) human–computer interaction (HCI) research in agricultural settings.

Machine Learning Approaches One direction of work focuses on building machine learning and deep learning models for different agricultural tasks. Saddami et al.²² explores CNN-based deep learning models for the task of automatically detecting leaf disease markers from pictures of rice plants. Their experiment results show that the best performing model achieved an accuracy of 99.8% in this task. Similarly, Guo et al.²³ presents a deep learning algorithm, PECAD, for accurate prediction of future produce prices based on past pricing and volume patterns. Based on experiment results, PECAD achieves ~25% lesser coefficient of variance than state-of-the-art baselines. Further, PLAN, a spatio-temporal deep learning algorithm, utilizes crowdsourced reports of locust observations for forecasting future migration patterns of locusts at high spatial and temporal resolution across East Africa.²⁴ Results indicate that PLAN achieves an AUC score of 0.9. Compared to several baseline models, it exhibits superior predictive performance on various locust movement forecasting tasks. In addition, Tabar et al.²⁵ leverages structural insights about temporal patterns of timeseries data to generate forecasts for three widely used crop-productivity related variables (actual evapotranspiration, reference evapotranspiration, and net primary production), which can be utilized by farmers to take corrective actions on their farms. CLIMATES outperforms baselines in a similar task and is evaluated by non-profit organizations (NGOs) for deployment evaluations. Bhardwaj et al.²⁶ designs a mechanism for procurement of agri-inputs based on volume discount auctions. Simulation experimentation show the efficacy of this mechanism. Bhardwaj et al.²⁷ proposes a deep learning

model to the crop prices prediction using various key determinants of crop prices. It produces a performance that is 20% better than the results available in the literature. A tool, CROP-S (CROP Planning System) is developed to determine an assignment of crop acreages to districts so as to maximize the profits of the farmers.²⁸ Through the world data on the top 7 crops grown in Karnataka, authors show that CROP-S are effective to increase the overall profits of the farmers. Patel et al.²⁹ designs the agricultural crop recommendation engine) to choose a crop or a portfolio of crops, to maximize the utility to the farmer. Through the publicly available data in India, this recommendation engine could effectively assist farmers in choosing crops. Nuru is a deep learning-based object detection model for diagnosing plant diseases and pests.³⁰ Through experiments, it could diagnose symptoms of cassava diseases at a higher accuracy than the agricultural extension agents and farmers. Turgut et al.³¹ utilizes Internet of Things (IoT), machine learning and explainable artificial intelligence (XAI) to build a crop recommendation system and improves the operational efficiency of crop cultivation based on weather and soil conditions. Although these works show promising performance across various agricultural tasks, they are often difficult to understand, deploy, or deliver to farmers via the Internet, which is not the case for all farmers. As a consequence, most these models remain at the stage of research prototype rather than becoming real-world products.

Generative AI Approaches Some recent work utilizes LLMs to improve pre-specified task performance on agricultural tasks. Tonmoy et al.³² proposes MobilePlantViT, a novel hybrid Vision Transformer (ViT) architecture designed for generalized plant disease classification. Extensive experiments across diverse plant disease datasets underscore the potential of automated plant disease classification in sustainable and resource-efficient smart agriculture systems. Kamangir et al.³³ introduces a multi-modal transformer (based on remote sensing imagery and short-term meteorological data) for pixel-level vineyard yield predictions. It excelled in spatial variability capture and yield prediction, particularly for extreme values in vineyards. Rakib et al.³⁴ creates a multi-modal approach for the task of soil moisture estimation. Extensive analysis shows that this approach achieves a Mean Absolute Percentage Error (MAPE) of 10.14%, outperforming traditional unimodal approaches. Other works focus on applying generative-AI techniques to

create interactive AI systems. For example, Samuel et al.³⁵ develops an AI-powered chatbot to enhance knowledge-sharing and education in agriculture using Large Language Models (LLMs) and a Retrieval-Augmented Generation (RAG) framework. The chatbot has achieved promising performance on four agricultural domains: agriculture and life sciences, agricultural management, agriculture and forestry, and agriculture business. Singh et al.³⁶ introduces Farmer Chat, a generative AI-powered chatbot to spread agricultural knowledge for smallholder farmers and rural users. It integrates voiced-based interactions, promoting more intuitive human-AI collaboration. Through real-world adoption, it improves farming practices, response quality, and user engagement for farmers. These works attempt to utilize Generative AI models to solve agriculture problems. However, these existing approaches are pull-only systems, they only provide answers to questions explicitly asked by farmers, and thus, no attempt is made to account for situations where farmers may not be aware that something in their agricultural practice needs immediate attention. In addition, response or feedback from farmers is not fully utilized in these existing approaches, which should ideally be leveraged to refine the predictions and recommendations made by these AI systems.

HCI Approaches Recent research has focused on user-centered tools that align with farmers' workflows to support decision-making. For example, engaging immigrant farmers in a co-design process to address challenges like outdated record-keeping and low literacy, which often lead to missed opportunities in crop disease management and budgeting.³⁷ Their findings indicate that culturally relevant, low-barrier interfaces can improve information management and decision support on farms. However, limited digital literacy in many rural communities remains a vital limitation to adopting such digital agriculture technologies. Arangurí et al.³⁸ indicates that factors such as age, education, and infrastructure strongly shape farmers' digital skills, which in turn correlate with technology. It also highlights that strategies—including contextualized training, locally tailored content, and peer learning—are effective in boosting digital literacy and, consequently, increasing the adoption of farming tool.

Among different types of attempts, one approach to bridge literacy and accessibility gaps is the use of voice-based systems. By delivering

expert advice via phone calls or voice interfaces, these systems make agricultural guidance available to farmers with minimal reliance on text. Dione et al.³⁹ demonstrates this in Uganda, where an interactive voice response (IVR) service provided pig farmers with biosecurity advisories. A field trial shows significant knowledge gains among IVR users, including those without prior training. Similarly, the Ama Krushi program (PxD's flagship digital agricultural advisory program) delivered weekly two-way voice messages and an on-demand hotline to tens of thousands of smallholders, offering region-specific advice on crops and livestock. The corresponding evaluation finds that voice-based advice helps farmers adopt improved practices, boosted yields, and reduced severe crop losses.⁴⁰ These deployments underscore the value of voice interfaces as scalable, user-friendly channels for agricultural knowledge, effectively complementing conventional extension by accommodating low-literacy users and providing timely decision support at scale.

3 Leveraging Large Language Models to Understand and Bridge Farmer Needs

Traditional AI systems have historically lacked high levels of interactivity.⁴¹ Using these systems, understanding their outputs, and deploying their recommendations in real-world contexts often requires significant technical expertise. This lack of accessibility has rendered such AI tools out of reach for low-resource populations, such as farmers, who may not have the technical skills or computational infrastructure needed to leverage these tools effectively.

Large language models (LLMs) and other recent developments in generative AI have created exciting opportunities for enhancing agricultural operations through improved communication, information sharing, and individualized assistance.^{42,43} Two key opportunities stand out: the ability of LLMs to understand and interpret farmers' needs, and their capacity to serve as intuitive interfaces that translate complex technical insights into actionable knowledge for non-technical users.

3.1 Understanding Farmers' Needs through Natural Interaction

One challenge in agricultural innovations is to effectively identify the needs, concerns, and problems faced by farmers, especially those in resource-constrained areas. This ground

viewpoint is often missed by traditional top-down AI techniques for agriculture, which results in solutions that are neither relevant nor widely adopted within real-world agricultural setups.

LLMs offer a novel perspective to reverse this dynamic by collecting dialogue data through a bottom-up approach. Through conversations in natural language, LLMs can engage directly with farmers in their native languages and dialects.⁴² This process gives farmers additional opportunities to express their concerns in their own languages. Through sufficient fine-tuning or domain adaptation on such data, LLMs could extract urgent needs and region-specific issues. In addition to facilitating data collecting, this capability makes agricultural innovation more collaborative by treating farmers as co-creators rather than passive consumers of technology.

3.2 LLMs as Human-Centric Interfaces for Knowledge Transmission

Beyond the need of understanding farmers' concerns, LLMs are essential in bridging the gap between the outputs of heavy-duty agricultural AI algorithms and their practical on-the-ground implementation by farmers.⁴⁴ Agronomic techniques, such as irrigation scheduling, disease modeling, and genotype-environment interactions, are frequently obscured by scientific language or digital literacy to access and utilize. For many farmers, this is a major obstacle, particularly in settings with limited technical background or connectivity.

LLMs can mitigate this issue by acting as an intelligent interface which converts complicated scientific results into understandable and actionable advice.⁴⁵ For example, after analyzing satellite data and weather prediction models, an LLM could transform such information into a customized message: "Based on weather trends and the location of the field, it's better to irrigate your wheat crop early tomorrow morning." This output can be transferred into local languages (through text or speech), allowing for greater accessibility and usability.

Additionally, LLMs are able to modify their final outputs (style of preference or explanations) according to user profiles. For instance, an experienced farmer may receive higher-level summaries and comparison recommendations, whereas in comparison, a novice farmer might receive detailed instructions with straightforward expressions of language and technical jargon. This adaptability positions LLMs as a flexible communication bridge, capable of aligning the outputs

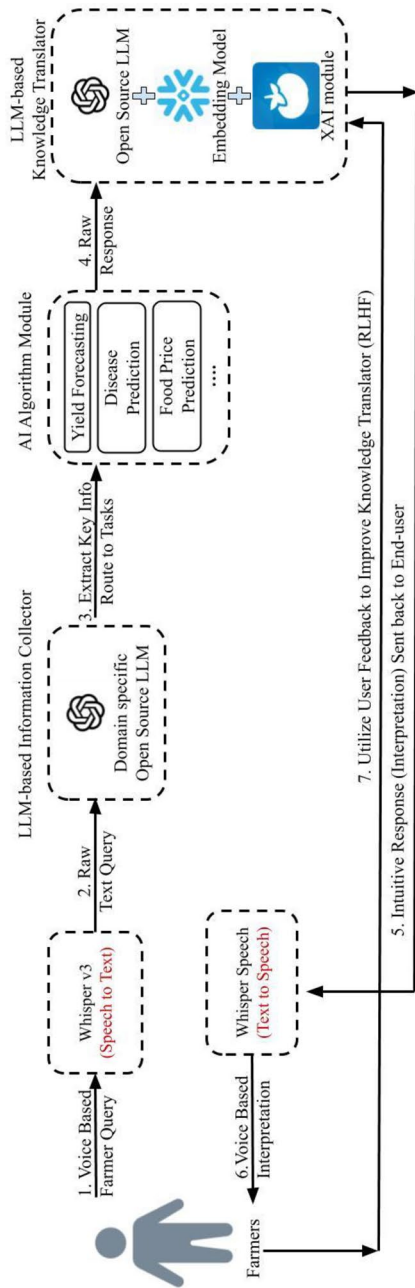


Figure 1: Voice-based agricultural AI service through generative AI.

of sophisticated models with the real-world needs and preferences of diverse agricultural communities.⁴⁶

4 Using Large Language Models to Rethink Farmer-AI Communication

In this paper, we argue that LLMs (specifically instruction-tuned models) can be used to rethink and redesign farmer-AI communication. As mentioned in the Introduction section, farmers are not tech-savvy, and the current AI algorithm based tools/services for agriculture are not designed to support such users with low technical literacy.⁴⁷ In addition, even though the predictions returned by AI algorithms may seem counterintuitive at times, farmers are not provided with any intuitive explanations for its predictions. Even though the field of Explainable AI (XAI)^{48–50} has worked on developing algorithms that can help resolve this issue, the explanations generated by XAI algorithms still need to be presented to the end users in an intuitive and user-friendly manner that they are comfortable with.⁵¹ Thus, the modality of interaction matters significantly. To foster better acceptance and usability among farming communities, it is essential to implement intuitive voice-based user interfaces that are more reflective of the farmers' everyday experiences. With this context in mind, we present an ideal (yet, easily achievable) vision of how LLMs could redefine farmer-AI communication.

This voice-based agricultural AI service (shown in Fig. 1) consists of five components: (1) a speech-to-text (speech recognition) module; (2) an LLM-based information collector; (3) an AI algorithms module; (4) an LLM-based knowledge translator (interpreter); and finally, (5) a text-to-speech (TTS) module. In addition, this AI service should also consider how LLMs can be utilized to generate safe and reliable responses, especially with farmers' unique preferences and constraints in mind. In the paragraphs that follow, we will discuss each of these modules in detail.

4.1 Speech Recognition Module to Collect Information from Farmers

In this AI service, an off-the-shelf speech recognition module can serve as a crucial first-stage interface for natural, low-barrier data collection. For many farmers in rural or resource-limited regions, they may lack digital literacy or access to text-based input methods. Through the voice-based interactions in local languages and dialects, the speech recognition module can collect

essential information about the farmers' profile and condition in a more inclusive and accessible manner.⁵² These speech signals are transcribed into structured raw input which serves as input for an LLM-based information collector (extractor) and downstream AI algorithms modules focused on decision support and resource optimization. Importantly, this module reduces the entry barriers and promotes continuous interactions between farmers and the AI service, forming the foundation for a more personalized AI agricultural service. OpenAI Whisper v3 can serve as the starting point to realize the functionality of this speech recognition module (although there are challenges that need to be tackled, as we discuss in Sect. 5).

4.2 LLM-based Information Collector

The LLM-based information collector module serves as a bridge between advanced AI agriculture algorithms and real-world natural language queries (and responses). This component serves as a natural language interface which is capable of understanding farmers' diverse and unstructured queries, translating natural language input into actionable machine-understandable tasks, and ultimately supporting informed decision-making in agriculture. AgriLLM⁵³ or AgriGPT⁵⁴ could be considered as the base model for this information collector. To be more specific, this module realizes four roles:

Understand and Process Farmers' Needs It serves as a natural language interface capable of interpreting and understanding the needs expressed by farmers in their own terms. Farmers often describe their problems using informal, region-specific, or domain-specific language, which may not align with the structured input requirements of typical AI systems. This module bridges this gap by interpreting natural language input and mapping it to specific agricultural issues such as pest outbreaks, crop disease, irrigation challenges, or market concerns. By fine-tuning the open-source LLMs on domain-specific corpora (such as agricultural regional reports and annotated farmer queries), the module (LLM itself) can better capture nuances, local terminologies, and intent.^{55,56}

Storing Contextual Memory Understanding specific questions and preserving context during interactions are essential for effective communication with farmers. The LLM-based information collector could incorporate a memory component

to retain relevant background information over time—such as the farmer's location, crop type and previous problems. The module can reply to follow-up inquiries and modify suggestions due to this contextual memory (or conversation history).⁵⁷

Extract and Structure Key Information This is the heart of our AI agricultural service. Once the farmer's query is received and processed, the LLM must convert unstructured meta messages into structured numeric inputs that downstream AI algorithms can utilize as input. This involves extracting key entities (e.g., crop type, symptoms, location), inferring missing information, and further processing the data into predefined schema which can be used as the inputs of different AI prediction (or resource allocation) tasks. The LLM can leverage prompt engineering or fine-tuned models to perform information extraction with high accuracy,⁵⁸ even in the presence of linguistic ambiguity or incomplete descriptions.

Route Query to Different Agricultural Prediction Models The final role of this information collector is to intelligently route the structured query and key information to the appropriate AI submodule. By identifying if the question relates to different tasks, the module serves as a semantic router and sends it appropriately.^{59,60} For instance, a crop disease-related query may be forwarded to a vision-based model if accompanied by an image, or to a rule-based knowledge model if it is text-only. The AI service can extend flexibly across several tasks because of this orchestration feature, all the while keeping the end-user interface consistent.

4.3 AI Algorithms Module

The AI algorithm module serves the computational core of this AI service. It's designed to support the arrangement of predictive and decision support tasks in agricultural domains. This module can store and integrate state-of-the-art machine learning and deep learning models to analyze multimodal agricultural data,⁶¹ such as satellite images, weather records and soil composition. Each predictive task inside the module can play the role of a service component, enabling modularity and extensibility. For example, one task may focus on the long-term yield forecasting which may require a specific AI algorithm,⁶² while another task concentrating on disease prediction may require a different AI algorithm.⁶³

This module will effectively store implementations of all these AI algorithms on a cloud-server setup, and these algorithms will be invoked by the LLM depending on the particular query being asked by the farmer.

While this module performs the core step for computational analysis, its full potential is unlocked when combined with the LLM-based interpreter which acts as a mediator between intricate AI outputs and farmers' everyday realities. It serves as a vital backend engine that enables the AI service to provide easily accessible, contextually aware, and human-centered agricultural assistance.

4.4 LLM-based Knowledge Translator (Interpreter)

One persistent challenge with the growing use of AI technologies in agriculture is how to communicate the results of heavy-duty AI algorithms to farmers through an intuitive and understandable interface. Despite the excellent performance of many AI-driven agricultural solutions (such as predictive modeling^{64,65} and decision support^{66,67}), these results are always expressed in abstract representations or technical jargon that are not meaningful to most low-resource farmers around the world.

To address this, the LLM-based knowledge translator module can serve as a core component in our AI service architecture. This module functions as an interface layer that transforms complex algorithmic outputs into intuitive explanations tailored to the end user's background and literacy level.⁶⁸ Leveraging the capabilities of large language models to understand and generate human-like text, this component ensures that farmers not only receive actionable insights but also understand the reasoning behind them. This module can be instantiated through three mechanisms.

LLM's Reasoning Ability LLMs have shown strong performance in contextual reasoning and multi-modal understanding.^{69,70} Depending on the local languages, dialects, and farmer knowledge levels, these models can be utilized (through fine-tuning or prompting) to interpret the raw output from AI algorithm module into a conversational or story-format final output. For example, commercial models (such as OpenAI o1 model⁷¹ and Anthropic's Claude 4 model) and open-sourced models (Qwen3 series⁷²) are shown

promising performance in solving and understanding complex tasks

Explainable AI (XAI) Through integrating the existing explainability frameworks (e.g., SHAP,⁷³ LIME⁷⁴) with the LLM, we could first generate structured explanations which are further interpreted and narrated by LLMs.^{31,75} In addition, the global explanation and local explanation of the AI model can also be utilized to explain AI output. For example, model distillation, a global explanation method, uses an interpretable model to mimic the black-box model's behavior by treating the black-box model as teachers so that the complex model is "trimmed" into a simple interpretable model.⁷⁶ Feature-importance explanation, which is considered as local explanation method, measures how important a particular feature of the predictive model.⁷⁷ What's more, counterfactual explanation offers a contrastive case that provides an opposite outcome with changes in the input.⁷⁸ All these approaches ensure accessibility, enabling the system to inform farmers about the rationale behind the methods by which the AI algorithm module makes decisions.

Retrieval-Augmented Generation (RAG) To ensure factual grounding and contextual relevance, the LLM can be coupled with a domain-specific retrieval system. We argue that the RAG pipeline framework can be used for an agriculture and farmer-specific translator, which would be capable of generating natural language responses to farmer inquiries.³⁶ In this case, using RAG will allow the LLM to access a diverse range of agricultural data sources, including extensive databases of agricultural information (that are publicly available online), expert knowledge, and past user interactions.³⁵ This RAG pipeline would be crucial to create an intuitive response, ensuring that the outputs generated from the AI-algorithm module are enriched and explained by agricultural domain data and tailored to individual farmer needs.

To construct a robust RAG pipeline, two primary components need to be designed: an embedding model and an instruction-tuned Large Language Model (LLM). Since there exists many possible off-the-shelf embedding models and instruction-tuned LLMs, it is necessary to mix and match different embedding models with different instruction-tuned LLMs to find the optimal combination. For example, as a starting point, it is reasonable to mix-and-match popular embedding models from the Massive Text

Embedding Benchmark (MTEB)⁷⁹ with powerful instruction-tuned LLMs from the UC Berkeley LMSYS Leaderboard.⁸⁰ For embedding model, we set up the snowflake-arctic-embedding and the nomic-embedding⁸¹ models (both known for excellent performance) as a starting point to convert agricultural knowledge (containing agricultural papers, books, reports) into embeddings into the vector database. For the instruction-tuned LLMs, Qwen3 series⁷² could be utilized as the initial choice to answer farmer specific queries into a context-aware manner.

4.5 Text-to-Speech (TTS) Module to Create Voice-based Response

In the last stage, we integrate a text-to-speech (TTS) module into this AI service to directly generate voice response. To be more specific, the natural language explanations of AI-driven insights post-processed by LLMs are further converted into the speech signals in local language or dialect of farmers. This design transforms the LLM into a voice-interactive interface, allowing farmers to receive complicated and detailed guidance or explanation through natural spoken language. This module promotes inclusivity and usability among people with low literacy or little expertise with digital technologies while reducing the dependency and limitation on text-based interface. This module can be realized through Whisper Speech API or open-source TTS models, such as Fish-speech⁸² or Chat-TTS.

4.6 Ensure LLM and Data Safety

LLMs can reflect and even amplify societal biases and stereotypes embedded in their training data, leading to discriminatory or offensive outputs.^{83,84} Additionally, these models may generate inaccurate, misleading, or false information, which can be particularly problematic when used in critical contexts like agriculture. When deploying LLMs with farming communities, it is crucial to put in place safety mechanisms that can ensure that the generated LLM outputs do not harm the end-users. The safety and reliability of the responses generated by the LLM can be realized through uncertainty quantification⁸⁵ and human-in-the-loop approaches.⁸⁶ To be more specific, the uncertainty quantification provides a way to measure the uncertainty/confidence of LLM outputs, allowing the system to identify instances where the model's responses may be less accurate or require additional validation. In such cases, implementing a human-in-the-loop approach involves leveraging domain experts (i.e., farmers)

to review and correct uncertain responses, thereby enhancing the safety and trustworthiness of the LLMs' output. Incorporating these safety measures ensures a more reliable and user-friendly AI-powered service for farmers, reducing the risk of incorrect or misleading information and building greater confidence among.

4.7 Create Preferred Response Through LLM Alignment

To ensure that the LLM used in this service generates responses that resonate with the insights and practical wisdom, we leverage farmers to collect feedback on LLM-generated responses, using it to reinforce desired behavior in the model.⁸⁷ Through this process, the LLM-based knowledge translator can learn to respond to farmer queries in a way that is more accurate, trustworthy, and contextually appropriate, leading to greater acceptance and trust among the farming community.

Reinforcement Learning with Human Feedback (RLHF) has emerged as a powerful method for fine-tuning Large Language Models (LLMs) by integrating human feedback into the training process.⁸⁸ This approach is commonly used to align LLM outputs with desired behavior, ensuring that model responses meet specific quality, accuracy, and context requirements.⁸⁹ Two major approaches for Reinforcement Learning with Human Feedback (RLHF): Proximal Policy Optimization (PPO)⁹⁰ and Direct Preference Optimization (DPO)⁹¹ could be utilized. Both approaches aim to fine-tune the Large Language Model (LLM) by incorporating human feedback, aligning the model's outputs with expert farmer knowledge.

To generate the human feedback dataset, each farmer query in this AI service will be processed by the LLM-based knowledge translator to produce different responses to ensure a range of outputs. These responses will be routed to the volunteer farmers who will annotate which response they find most accurate, useful, and contextually appropriate. These annotations will form the basis of the human feedback dataset, which will then be used to implement the RLHF techniques.

5 Challenges of Applying Generative AI in Real World Agriculture Scenario

Despite the promising potential of generative AI in transforming agriculture in the global world, several critical challenges impede its effective deployment and adoption. These challenges

span linguistic, infrastructural, and data-centric dimensions.

5.1 Linguistic Diversity

The first challenge to deploy such LLM powered AI service for agriculture lies in global linguistic diversity. There are more than 7000 known living languages in the world.^{92,93} What's more, many of them are regional dialects and are expressed as code-mixing.⁹⁴⁻⁹⁶ Based on this situation, farmers tend to communicate in their native languages, which can differ significantly across the areas-even within the same country. A significant barrier to the creation of inclusive generative AI services that can efficiently assist all farmers is this linguistic fragmentation. Large language models (LLMs) and other cutting-edge generative AI models are typically trained on high-resource languages like English and Chinese.⁹⁷ Their utility and accessibility are restricted by this training approach for farmers who speak regional languages. Furthermore, dialectal differences can affect understanding and confidence in AI-generated information even within a single country (especially linguistically diverse countries like India).

Moreover, even when translation or multilingual capabilities are integrated, nuances in agricultural vocabulary, cultural expressions, and idiomatic usage may not be accurately captured.⁹⁸ Poor handling of local agriculture terminology might result in misinformation or a decline in trust in the AI services because they frequently have no direct equivalents in other languages. Examples of these terms could be meteorological phenomena, farming methods, diseases, and pests. Developing robust multi-lingual support for such AI services requires not only large-scale annotated corpora for these languages and dialects but also the knowledge to adapt AI generated responses to agricultural practices unique to a certain region.

RLHF provides a potential solution for this linguistic fragmentation issue. By incorporating feedback from local speakers (including farmers and agricultural experts), the RLHF procedure could fine-tune LLMs to better understand and generate content in regional languages and dialects. For example, Zhang et al.⁹⁵ proves that LLMs could better handle code-mixing through a data and cost-effective way based on machine translation task. Through the combination of multilingual pretraining and data collection efforts in target languages, RLHF can significantly enhance the accessibility and effectiveness

of generative AI tools for agricultural users in diverse linguistic settings.

5.2 Limited Access to Agriculture and Farmer Data

The restricted availability and accessibility of agricultural and farmer-specific data is another significant obstacle. Generative AI systems rely heavily on structured and unstructured domain-specific data for fine-tuning, post-training, and validation to produce relevant, context-aware outputs.⁹⁹ However, in the global agricultural context, data is often sparse, fragmented, or inaccessible.¹⁰⁰ For example, historical weather records, soil health data, and even individual farmer profiles are inconsistently maintained across regions. Even existing data is often incomplete or out of date. Building AI services that can give farmers region-specific, individualized, and up-to-date recommendations is challenging without well-maintained data.

Privacy and consent issues further bring difficulties to access to farmer-level data.¹⁰¹ Farmers may be reluctant to share personal or daily operational records due to mistrust of AI services or lack of digital literacy. In rural areas, the transparent data governance procedures and ethical data gathering process are still in their infancy.¹⁰² This limits the development of such AI services learn from real-world examples and transfer to the agricultural domain. Coordinated efforts in digitalization, data standardization, and the adoption of farmer-centric data policies will be necessary to meet this challenge and guarantee the ethical and use of data in agricultural generative AI applications.

The privacy obstacle can be overcome with federated learning (FL), a decentralized approach where multiple farms collaborate to train a shared ML model without centralizing their data.¹⁰³ FL enables collaborative model training across farms, addressing the issue of data scarcity by aggregating valuable context-specific data, preserving data privacy, and allowing for flexible participation and reduced communication. This approach fosters real-time decision making and adaptability to local conditions.

5.3 Limited Internet Penetration and Digital Infrastructure

Although the world has witnessed significant growth in digital connectivity over the past decade, the successful deployment of generative AI in agriculture is heavily dependent on internet accessibility. Even though the number of people

using smartphones has increased quickly, many farmers, particularly in remote or economically undeveloped areas, still lack reliable access to high-speed internet and smart devices.¹⁰⁴ This digital divide restricts their ability to interact with AI-powered services (for example, those services require cloud-based processing or real-time updates). Urban farmers, on the other hand, stand to gain much more from these developments, resulting in a digital gap within the agriculture industry.

One promising avenue is to integrate the voice-based interaction system. Unlike text-only interfaces which requires literacy or technical skills, voice-based techniques offer a more intuitive way for farmers to interact with AI services. Such systems could function on fundamental mobile phones or low-end smartphones, making them accessible even in areas with limited digital infrastructure. Through the support for local languages and dialects, voice-enabled system could deliver real-time services. This also encourages traditionally underrepresented farming communities to be included in the digital transformation of agriculture.

Realizing the full potential of generative and voice-enabled AI systems also requires substantial computational resources and real-time data processing. Generative AI models are inherently resource-intensive, often requiring large-scale computation for training and inference. To manage this agricultural context (where connectivity and power supply can be inconsistent), edge-cloud hybrid infrastructures offer a practical solution.¹⁰⁵ By performing lightweight inference or adaptation on edge devices such as local servers, or mobile gateways, and reserving complex generative reasoning tasks for the cloud, systems can maintain responsiveness while reducing dependence on continuous internet access.¹⁰⁶

5.4 Misalignment Between AI Research and Farmers' Real-World Needs

There is a significant gap between the real-world, practical needs of farmers and the development of AI tools in research settings. It's possible that AI researchers aren't directly aware of the realities and daily struggles encountered by rural farmers. As a result, many AI models or papers are targeted to the ideal or assumed need which do not reflect the context of the global agricultural development.^{107, 108} This misalignment leads to the creation of AI models or systems that may be technologically impressive but lack relevance or usability for farmers.

Furthermore, farmers frequently emphasize immediate and concrete concerns (such as seed quality, market price).¹⁰⁹ While AI systems or models prioritize optimizing yield projections without an interactive loop with farmers or agricultural issues in the real world. Without continuous engagement with agricultural communities, AI systems may not evolve in a direction that benefits their intended users.

Bridging this gap requires interdisciplinary collaboration not only from data scientists but also from local NGOs and the farmers themselves in the co-design process. This approach ensures that AI models are informed by reality on the ground and are better tailored to local contexts. NGOs frequently play a crucial role as mediators since they owe trust from local farmers, are aware of community dynamics, and can assist in transforming technical initiatives into acceptable behaviors. Similarly, including farmers in this process help the AI system better reflect their priorities and concerns, enhancing long-term sustainability.

6 Conclusion

In this paper, we explore the potential of generative LLMs as a transformative interface between the agricultural community, low-resource farmers and increasingly complex agricultural AI tools. Through the interactive capabilities of these models, we argue that they present an appropriate path for farmers to access cutting-edge agricultural decision-support systems. Unlike conventional user interfaces or AI tools, Generative AI models offer a conversational modality that responds to user intents and is flexible and intuitive to use even for users with low technical literacy.

As these LLMs continue to evolve their information extraction, understanding, and reasoning abilities, their integration into agricultural workflows could fundamentally reshape how knowledge is shared, decisions are made, and productivity is scaled sustainably. This work's broader implications could go beyond agriculture. They could serve as general-purpose mediators in a variety of fields with information asymmetry and limited infrastructure.

However, realizing this vision will require future research to solve several challenges, which include but not limited to ensure domain-specific robustness, mitigate risks of LLMs' hallucination, and optimize training strategy of LLMs to take into account language and worldwide agricultural variety. Furthermore, developing systems that are

both technologically sound and socially anchored will require cooperation between AI researchers, domain experts, HCI specialists, and rural people.

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Declarations

Conflict of Interest

Regarding the content of this work, the author affirms that there are no conflicts of interest. There were no financial or commercial ties that might be seen as a potential conflict of interest when the research was being done. The conception, execution, and interpretation of this work were not influenced by any affiliations, financial interests, or personal or professional relationships.

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